

openvaf-r — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

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openvaf-r — Full Change Report

Every modification applied to the OpenVAF-Reloaded compiler in this project, with its reason. The baseline is the pristine OpenVAF-Reloaded source tree (as vendored in the project's `original/` snapshot); the current state is the tree committed in this repository under `OpenVAF-master-20260610/`. Each entry links the enhancement write-up in [enhancements_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [ngspice_changes_full-report.md](#) covers the simulator side.

Maintenance note: this report is updated whenever an enhancement touches compiler sources. The per-enhancement index at the end tells you the last change it covers.

Scope summary. ~120 modified source files across the compiler pipeline, two new files (`hir/src/elaborate.rs` — the elaboration pass — and `hir_def/src/item_tree/diagnostics.rs`), plus three classes handled wholesale: the `test_data/` fixtures and snapshots (95 files — new test inputs and snapshots regenerated whenever a feature legitimately changed descriptor contents), the `integration_tests/*/*.osdi` reference objects (regenerated at each ABI bump), and the removal of the AGPL vacask submodule. The compiler builds warning-free; every change below was regression-gated by the example-suite arsenal, the integration suite, and the `VA_TEST` industry-model corpus.

1. Lexer — `openvaf/lexer/` (`lib.rs`, `cursor.rs`)

- Based integer literals (`8'hFF`, `'sb101`): the lexer had only a commented sketch; underscores crashed it. Both the sized and unsized forms now lex, requiring a digit after the base (a silent-0 trap otherwise) ([E-46](#)).
- Don't-care digits `x/X/z/Z/?` accepted in the binary/octal/hex digit runs for `casex/casez` items ([E-78](#)).
- Escaped identifiers (`\my-net!`) lexed per the LRM ([E-46](#)).
- `// comment at EOF` hang fixed: with no trailing newline the line-comment loop spun forever on the EOF character — the only EOF-unsafe loop in the lexer ([E-35](#)).

2. Tokens and grammar — `openvaf/tokens/`, `openvaf/parser/`, `veriloga.ungram`, `sourcegen/`

`tokens/src/parser/generated.rs` gains the keywords and node kinds each language feature needed, threaded through `parser/src/grammar/*` (`stmts.rs`, `expressions.rs`, `items.rs`, `items/module.rs`), the grammar source of truth `veriloga.ungram`, and the `sourcegen/` generators (`ast/src.rs`, `hir_builtins.rs`, `mir_instructions.rs` — the templates the generated token/AST/builtin/instruction tables come from):

- `do ... while` post-test loops ([E-19](#));
- `paramset/endparamset` blocks ([E-21](#));
- `{...}` concatenation and `{n{...}}` replication as real operators, with `'{...}` remaining the typed aggregate ([E-34](#));
- array-literal expressions actually parsed (they were fully scaffolded but unreachable) ([E-4](#), [E-14](#));
- vectored (bus) net declarations and bit-selects ([E-3](#)), multi-index bit-selects for N-D arrays ([E-15](#)), and the LRM name-then-range declaration order ([E-18](#));
- module instantiation items ([E-5](#)), `generate for/genvar` ([E-8](#)), and `generate if/generate case` with optional block labels (previously mis-parsed into a broken `GENERATE_FOR`; [E-67](#));
- `defparam` — a reserved word with no grammar rule ([E-58](#));
- event OR lists `@(cross(...)` or `timer(...)`) via a new or keyword and looped event grammar ([E-59](#));
- `casex/casez` sharing the case rule, the keyword token carrying the flavor ([E-78](#));

- `@(initial_step)/@(final_step)` with analysis-phase lists (E-7, E-53);
- indirect branch assignment `V(x): V(y) == expr;` (E-2);
- the `<</>>>` arithmetic-shift tokens (plus a pre-existing lexer bug fix) (E-6);
- hierarchical branch reference tails `(.branch(a,b) / .branch(<p>))` swallowed into the path node so the enclosing item's CST stays whole for the elaboration rewrite (a keyword in path position used to shred the item, leaving the hole scanner truncated text) (E-86);
- part-selects in instance connections `(inst (out[3:2], in))`: the bit-select bracket accepts an optional `:` `expr`, with the colon token in the CST distinguishing a range from multi-dimensional indexing — no new node kind (E-85);
- panic-free direction parsing: `port_decl/func_arg` asserted the next token was a direction (`bump_ts`), which any non-Verilog text reaching a module-head port list turned into a compiler crash; both now emit a diagnostic with one token of forced progress (E-84);
- operator precedence corrected against LRM Table 4-2: `%` bound tighter than `*/` (so `6*7%4` evaluated to 18, not 2), and `xnor` split from `xor` (E-38);
- derived-nature parents emitted the wrong node kind (`NAME_REF` where the AST wanted `Path`), leaving the entire inheritance machinery unreachable (E-39).

3. Preprocessor — **openvaf/preprocessor/** (+ **basedb** diagnostics)

- Ten compiler directives that previously hard-failed as undefined macros: ``default_discipline`, ``celldefine/`endcelldefine`, ``unconnected_drive/`nounconnected_drive`, ``timescale`, ``line`, ``pragma`, ``undefineall`, ``default_nettype` (E-6).
- ``default_transition` parsed and threaded to `transition()` lowering (E-47).
- Recursive macro expansion crashed the compiler (stack overflow, direct and mutual): the `MacroRecursion` diagnostic existed but was never emitted, and its report renderer was a `todo!()`. The guard pushes around the macro *body* expansion only — an entry-scoped guard false-positives on the legal `QUAD(x) = TWICE(TWICE(x))` nesting (E-65).

4. Syntax / AST — **openvaf/syntax/** (**ast.rs**, **generated/nodes.rs**, **expr_ext.rs**, **node_ext.rs**, **name.rs**, **validation.rs**, **error.rs**)

- Typed AST nodes for every new construct above (generated from the ungrammar).
- Literal decoding (`expr_ext.rs`): based-int parsing with size/sign handling (E-46) and the mask-aware variant returning `(value, x_mask, z_mask)` for `casex/casez` (E-78); a compiler crash on large bare-integer-shaped literals fixed by falling back to a float constant (E-4).
- String unescaping rewritten as a single pass: the sequential `str::replace` chain corrupted overlapping escapes (`\\n`) and octal `\\ddd` was missing; all consumers route through one function (E-48).
- **Name::resolve** ate the last character of escaped identifiers (the committed `std.va` snapshot had baked the bug in as `logi`) (E-46).

5. Elaboration — **openvaf/hir/src/elaborate.rs** (new file)

The compile-time flattening pass: every stage downstream of the parser is architected around exactly one flat module per artifact, so hierarchy is resolved by recursively inlining instantiated modules — alpha-renamed per instance, ports bound to the caller's nets, parameters bound to overrides — into an ordinary flat module *before* the rest of the pipeline runs. Built for module instantiation (E-5) and grown into the home of every structural feature:

- `generate for/genvar` unrolling (E-8), nested loops, anonymous blocks, `generate if/case` with elaboration- time constant folding, and the `genvar`-substitution fix (`genvars` in ordinary expressions were routed through identifier renaming, which re-escaped `1e3*(i+1)` into a broken escaped identifier; they now become literal-value holes) (E-67);

- implicit nets from undeclared instance connections, discipline derived from the connected port, prefixed per instance so no cross-instance shorts (E-41);
- hierarchical names `u1.u2.node` and `$root` via an instance-chain map and a hole scanner (E-49);
- **defparam** resolved through the same chain rewrite, with the LRM-mandated precedence over instance `#()` overrides (E-58);
- net concatenation in port connections expanded bit-by-bit onto vectored ports (E-59);
- bus slicing onto instance arrays and bus ports (E-5); paramset twin-module rendering (E-21); net initializers preserved through re-rendering (E-45); escaped identifiers re-rendered correctly (E-46);
- undefined instantiation targets are a hard error — they used to be silently dropped from the rendered output, turning a typo'd module name into an invisible open circuit; discipline-named misparses (`electrical out [0:2]`; parses as an instantiation) and paramsets dropped over unresolvable targets get tailored messages (E-84);
- ``__FILE__`/ ``__LINE__` expanded by a textual pre-pass run before the other passes (preprocessor tokens are (kind, span) pairs into existing source and cannot carry synthesized literals): ``__FILE__` becomes the root file's basename (machine-portable provenance, the E-58 rule), ``__LINE__` the exact 1-based line; string/comment occurrences are skipped and replacements are inline so later line numbers stay true (E-85);
- part-select actuals sliced onto bus ports in `bind_port` (`base[msb:lsb]`, constant bounds; positional and named forms; width-1 slices onto scalar ports degrade to the bit-select) with the same ascending bit-order convention as full-bus slicing (E-85);
- hierarchical branch probes (E-86): the top module's absolute chain map rides into every inlined child so SIBLING bodies resolve `V(top.a1.b)/$root...` references; unnamed-branch forms expand to the flattened node pair; port-branch probes `I(inst.branch(<p>))` read a 0V ammeter synthesized at flattening (which also fixes child-declared branch (<p>) port branches, broken after inlining); hierarchy-bound monitor modules are omitted from standalone output; ground/net-type declarations in inlined children keep their keyword;
- **\$port_connected** resolved at flattening time to a literal `(1)/(0)` per instance — after inlining, an open port is just a synthesized local net, so the builtin used to fail validation in exactly the unconnected case it exists to detect; top-level modules keep the native OSDI connected-flag path (E-84).

6. Definition layer — **openvaf/hir_def/** (+ **openvaf/hir/** mirrors)

`body.rs/body/lower.rs/body/pretty.rs, expr.rs, item_tree*, namer.rs, builtin.rs, data.rs, types.rs, db.rs`, plus the new `item_tree/diagnostics.rs` and the `hir` façade (`body.rs, lib.rs, db.rs, diagnostics.rs`):

- statement/expression collection for every new construct: `do...while` (E-19), repeat and disable (E-9), event OR lists (E-59), concat/ replication (E-34), array aggregates and whole-array assignment (E-14), `CaseKind` + per-item don't-care masks with a stray-literal list (E-78);
- N-D array machinery: per-dimension size lists, multi-index bit-selects, per-element parameter expansion (E-15); declaration initializers on (N-D) arrays via a `Var::array_index` leaf split (E-43);
- **localparam** made genuinely non-overrideable (it behaved exactly like parameter) with derived localparams still tracking inputs (E-9); paramset lowering as twin modules whose bound params are localparams with override expressions (E-21);
- name resolution: `electrical ground gnd`; ordering (E-9), nature-attribute access (`net.potential.abstol` — the third "scaffolded-but-unwired resolver boundary" found) (E-45), derived-nature resolution (E-39);
- **builtin.rs** (the system-function registry): the noise-table pair (E-9), the full `$random/$dist_*/$rdist_*` family (E-10), file I/O and string functions (E-11), the last fallback group (`$simprobe, aliases, plusargs`) — after which `is_unsupported()` is empty (E-12),

`$simparam$str` (E-25), `$realtime` (E-59), `$table_model` (E-16);

- new diagnostics file: wrong-arity calls produced invalid HIR and crashed downstream — now a proper diagnostic (functions and parameters both) (E-43);
- multiple analog blocks collect into `entry_stmts` in source order — the as-if-concatenated LRM semantics, by construction (E-60).

7. Types and validation — `openvaf/hir_ty/`

`inference.rs` (+ `inference/fmt_parser.rs`), `builtin.rs` + `builtin/generated.rs` (signature tables), `lower.rs`, `types.rs`, `validation.rs`, `validation/body.rs`, `diagnostics.rs`:

- signature tables — a recurring defect source: `ANALYSIS` made variadic (E-30); `$table_model` given shape-synthesized varargs signatures for any dimension (a generic-varargs fallthrough *truncated* multi-arity signature lists — varargs builtins with signature lists need their own inference arm) (E-40); `TRANSITION`'s table was one argument short (3-argument calls crashed, 4-argument worked by accident) (E-47); `transition()` input typed `Real`, a `$dist` typo fixed (E-49); `SIMPARAM_STR` returned `Real` instead of `String` (E-25);
- format-string inference (`fmt_parser.rs`): terminates on every conversion character and reports it, so `%5d/%-8s/%08x` type their arguments correctly — previously only real conversions accepted flags/width (E-71);
- array inference: element-wise array case, array-literal function arguments (which previously bound nothing and silently returned 0), integer arrays typed `Real`, output-literal args accepted silently — all fixed (E-33); whole-array function arguments/outputs/returns (E-18, E-20, E-23); untyped function args default to `Real` instead of an ICE masquerading as an initializer bug (E-43);
- **Type: :base_type** infinite loop on any array-type check (it looped on `self` instead of the cursor) — hung the compiler on simple type mismatches (E-14);
- string handling: ternary over strings (`SELECT` + string binary ops appended to the operator tables) (E-37);
- validation: loop bodies get their own context so the analog-operator restriction reports "loops" (LRM 4.5.1), not "conditions" (E-70); call-graph cycles among analog functions are clean errors naming the cycle instead of a recursive-inliner stack overflow (E-59); `casex/casez` restrictions (stray don't-care literals, `x` digits under `casez`, non-integer discriminants) (E-78); domain discrete with natures rejected per LRM 3.6.2.2 (E-50); parameter defaults exempt from their own ranges (the CMC "feature disabled" idiom — stock rejected `diode_cmc`, BSIM-CMG, PSP-HV and the HiSIM family at setup) while given values stay fully validated (E-56);
- part-selects outside port connections diagnosed (the E-78 stray-list pattern: body lowering collects them, validation reports a dedicated error naming the supported connection form) (E-85);
- contributions to port branches diagnosed (`ContributeToPortFlow`, with the declaration site labeled): `I(pb) <+ ...` on a branch (`<p>`) `pb`; slipped through the write path unvalidated and panicked during lowering (E-84).

8. Lowering — `openvaf/hir_lower/`

`expr.rs`, `stmt.rs`, `ctx.rs`, `callbacks.rs`, `parameters.rs`, `state.rs`, `fmt.rs`, `lib.rs`:

- analog operators: `laplace_nd/np/zd/zp` via exact state-space realization (E-4) with complex pole/zero pairs per the LRM's (re, im) convention (E-31); `zi_*` via bilinear transform (E-6); `slew/transition` as a saturating tracking loop (E-6) with the LRM negative `max_neg_rate` convention honored (the sign defect made `slew` ignore its input and ramp away) (E-61) and a DC-singularity clamp via an `EnableIntegration` identity (E-47); `idtmod`'s modulo wrap moved off the reactive residual (it diverged) and an argument-index bug fixed (E-27); `idt` initial conditions survive into transient (E-28); the `idt` assert/reset forms with smooth reset dynamics (E-52); `limexp` kept stateless as a documented decision (E-13);

- **\$table_model**: 1-D piecewise-linear lowered to differentiable MIR so autodiff yields the Jacobian for free (E-16); N-D multilinear as recursive 1-D (E-17, E-40); natural cubic splines with the precomputed moment matrix (E-22);
- events: cross/above/timer with persistent previous-value slots (E-8); OR lists folded with a select-based bool-or (a raw ior ICEs const-eval) (E-59); final-step gating and phase-list AND-ing (E-53);
- variable persistence: genuine per-instance hidden_state storage behind a two-pass MIR build (E-7), typed slots so integer state stopped crashing instruction selection (E-32);
- callbacks(callbacks.rs, fmt.rs): the \$display family with the full [flags] [width] [.precision] surface and %h/%b/%r translation (E-71); file I/O and string formatting through a generalized PrintDst sink (E-11); the deterministic seed-and-salt RNG lowering (E-10); \$simplparam\$str (E-25); simulation-control return flags (E-55); \$discontinuity (E-24);
- operator fixes: unary ~ emitted arithmetic negate; the constant folder sign-extended >> where the runtime was unsigned — const and runtime disagreed (E-37);
- masked casex/casez comparisons (two iands ahead of the existing integer equality) (E-78); array function-argument writeback (E-20) and array returns via a function-named var-array (E-23);
- named port branches(branch (<p>) pb; — LRM 3.7.2): a flow probe of a PortFlow-kind branch routes through the same CurrentKind::Port param as a direct I(<p>), so E-29's defining equation covers both spellings; probing one used to hit unreachable!() (E-84);
- literal if conditions lower only the taken branch — a dead analog operator(transition()) under if ((0)), the exact shape the \$port_connected rewrite produces) survived const-folding as a detached-but-interned op whose state setup read optimized-away values and aborted codegen (E-84).

9. MIR core and optimization — **openvaf/mir***

mir/ (instructions.rs + instructions/generated.rs, dfg.rs with dfg/phis.rs and dfg/uses.rs, dominators.rs, flowgraph.rs with flowgraph/transversal.rs, layout.rs, cursor.rs, lib.rs, builder/generated.rs), mir_opt/ (simplify_cfg.rs, simplify.rs, const_eval.rs, global_value_numbering.rs), mir_autodiff/ (builder.rs, lib.rs, live_derivatives.rs), mir_llvm/ (builder.rs, intrinsics.rs), mir_build/, mir_interpret/:

- three pre-existing general CFG bugs, all found via event-counter verification: a dangling-reference bug in unreachable-block removal, a multi-exit post-dominance bug in the dominator-tree builder, and a block-merge bug that could corrupt which block the DAE builder treated as the exit (E-8);
- the post-dominator sink-root pathology that let dead-code elimination delete op-dependent \$fatal calls (side-effecting callbacks under op-dependent control now stay in eval; control dependence computed by arm-reachability) (E-55);
- split_tainted.rs tolerates layout-detached branch instructions (a branch whose condition const-folded has no CFG effect to taint; it used to unwrap and panic) (E-84);
- const-eval agreement with runtime semantics for shifts and the bitwise-not fix (E-37);
- autodiff: noise operators process before any ddt (a shared ddt evaluated between two noise operators silently dropped the second source), and the reactive coupling of a ddt-shaped noise wave is registered so small-signal pruning keeps it (E-54);
- **llvm.fabs.f64** registered in the LLVM intrinsics table — it never had been, and the signed noise-power convention needed it (E-42);
- new opcodes/instructions supporting the operator work (barrier/ integration-identity forms; MIR deliberately has no fabs, so lowering composes it) (E-47, E-52).

10. DAE construction — **openvaf/sim_back/**

dae.rs + dae/builder.rs, topology* (incl. linalize.rs, small_signal_network.rs), noise.rs, context.rs, lib.rs:

- implicit equations for indirect branch assignment (one new unknown
 - residual per statement) (E-2) and the absdelay synthetic two-node form (E-1) — with equation indices kept stable by mapping dead/collapsed unknowns to zero-contribution placeholders instead of renumbering;
- port-flow probes **I(<port>)**: a TODO stub returning 0 became a synthesized DAE unknown mirroring the node's KCL residual (E-29);
- probe-only branches: probing a never-contributed branch read 0 and an open circuit — a 0 V-source pass materializes them, enabling ideal ammeters and flow-only signal-flow disciplines (E-36);
- noise factors (noise.rs): equation-path noise was silently lost (never attached to the DAE); factors generalize to $re + j\omega \cdot im$ with no extra matrix unknowns (E-54); same-named source grouping metadata (E-42);
- idt reset-mode charge dynamics and conditional step bounding (E-52).
- voltage-source branches feeding internal nodes were open circuits at DC: the small-signal (noise/ac_stim) pruner keyed node registration on the branch's LIVE voltage unknown, so a pure $V(a, f) \leftarrow expr(V(a, f) \text{ never read})$ registered nothing and the node's conduction silently moved to the AC-only residual; any voltage-capable branch now disqualifies its nodes (E-86);
- probed **V(x, y) <+ 0** branches were node-collapsed away, making I(branch) read zero; hint pairs whose branch current is a DAE unknown are suppressed (the unprobed collapse idiom is untouched) and NodeCollapse::hint tolerates suppressed pairs; pinned by the permanent vsrc_internal_node snapshot test. Behavior change: a collapsible branch whose current the model references (MVSG_CMC's access resistances carry noise on flow(d, drc)) stays a real 0V source — electrically identical, two extra matrix rows (E-86).

11. OSDI code generation — **openvaf/osdi/**

lib.rs, metadata.rs + metadata/osdi_0_4.rs, inst_data.rs, eval.rs, load.rs, setup.rs, compilation_unit.rs, ndatable.rs, stdlib.c, build.rs, reference headers:

- descriptor emission for every ABI evolution (see the ngspice report's ABI table): the absdelay/last_crossing info arrays (E-1, E-6), Osdinode.nodeset (E-45), the ac_stim source array + load_ac_stim (E-51), stride-2 signed noise pairs in load_noise (E-54) — the version tuple lives in osdi/src/lib.rs;
- **eval.rs**: final-step flag gating (E-53); simulation-control return flags (E-55); the ac_stim large-signal value (was an unreachable!() panic) (E-26);
- **inst_data.rs**: absdelay slot layout (E-1); hidden-state slots typed by the variable (ty_f64 was hardcoded — integer state fed f64 loads into integer MIR ops and aborted instruction selection) (E-32);
- **compilation_unit.rs** (print codegen): the %b segfault — the binary-formatted string was remembered for free() but never pushed to the snprintf argument list, so the matching %s consumed a garbage pointer (E-71); the print-callback machinery with its shadowed-fun and volatile-table IPO traps (E-11);
- **stdlib.c**: the file-descriptor table behind \$fopen/\$fdisplay/... (E-11); the splitmix64 osdi_rng_* runtime (E-10); a simparam_str loop/return bug (E-25);
- **ndatable.rs**: ddt/idt natures resolved through resolve_nature_index instead of panicking on derived natures (E-39);
- **setup.rs**: range validation with default-exemption (E-56); a 64-line abandoned unsafe experiment (VoidAbortCallback) deleted in the warning cleanup (E-66);
- **build.rs**: the copied-target stdlib compilation trap fixed (E-10).

12. Driver, libraries, and infrastructure

- `openvaf-driver/src/main.rs`: hard errors now exit non-zero — the error arm printed the failure but fell through to a success exit, so every elaboration failure looked like a successful compile to shell scripts (a quirk first noticed during E-58's work) (E-84);
- `openvaf-driver/src/crash_report.rs`, `linker/src/lib.rs`, `lib/bforest/` (`map.rs`, `pool.rs`, `set.rs`), `lib/typed_indexmap/` (`set.rs`): the zero-warning cleanup ($44 \rightarrow 0$ across 13 crates: lifetime annotations, dead code, nightly cfgs, an unused LLVM-prefix probe) (E-66);
- `basedb/` (`ast_id_map.rs`, `lints.rs`, `lib.rs`, `diagnostics/preprocessor_error.rs`, `diagnostics/syntax_error.rs`): AST-id plumbing for elaboration-created items (the `AstId::from_erased` placeholder used by array returns; E-23), preprocessor/syntax diagnostic rendering for the new errors;
- `.llvm-version` pins the LLVM 18 toolchain expectation; the `hir` and `preprocessor` `Cargo.tomls` carry the dependency additions their new code needed.

13. The test suite — `openvaf/openvaf/tests/`, `lib/mini_harness/`, `test_data/`, `integration_tests/`

The fork's own integration suite over real compact models had never run in this project: its OSDI loader was frozen at ABI 0.4, its optional VACASK legs panicked on a never-initialized submodule, and the tests hide behind `RUN_DEV_TESTS=1`. Fixed test-side only (E-68):

- `tests/load/osdi_0_4.rs` + `load/mod.rs`: loader structs synced to ABI 0.7; `mock_sim/mod.rs`: stride-2 noise convention; `mini_harness`: missing directories skip with a note instead of panicking;
- VACASK removed entirely (`.gitmodules`, the submodule, 34 harness legs and their stale snapshots): AGPL-3.0 cannot be vendored here;
- `test_data/` (95 files as a class): new fixtures for new features and snapshots regenerated whenever descriptor contents legitimately changed — every regeneration reviewed (the diffs are the project's own features: `flow(<port>)` unknowns, probe-only branches, implicit-equation nodes); `integration_tests/**/*.osdi` regenerated at each ABI bump.

14. Per-enhancement index (compiler-touching enhancements)

Enhancement	Pipeline areas	One line
E-1	<code>sim_back</code> , <code>osdi</code>	<code>absdelay()</code> synthetic-node DAE + descriptor arrays
E-2	<code>parser</code> → <code>hir_lower</code> , <code>sim_back</code>	indirect branch assignment (implicit equations)
E-3	<code>parser</code> , <code>hir_def</code>	vectored (bus) nets with bit-selects
E-4	<code>hir_lower</code> , <code>syntax</code> , <code>parser</code>	<code>laplace_*</code> state-space + array-literal parsing
E-5	<code>elaborate.rs</code> (new), <code>parser</code>	module instantiation via compile-time flattening
E-6	<code>preprocessor</code> , <code>parser</code> , <code>hir_lower</code> , <code>osdi</code>	directives, shifts, slew/transition, <code>zi_*</code> , <code>last_crossing</code>
E-7	<code>hir_lower</code> (state), <code>osdi</code>	<code>@(initial_step)</code> gating + variable persistence
E-8	<code>parser</code> , <code>elaborate</code> , <code>hir_lower</code> , <code>mir/mir_opt</code>	generate for + events; three general CFG bug fixes
E-9	<code>hir_def</code> , <code>hir_lower</code> , <code>sim_back</code>	noise tables; <code>localparam/ground/strings</code> ; <code>repeat/disable</code>

Enhancement	Pipeline areas	One line
E-10	hir_lower, osdi stdlib	\$random/\$dist_* deterministic RNG
E-11	hir_lower, osdi	file I/O + string functions (PrintDst)
E-12	hir_def, hir_lower	last unsupported builtins as LRM fallbacks
E-13	hir_lower	limexp kept stateless (documented decision)
E-14	hir_def, hir_ty, hir_lower	array literals/params/dynamic indexing; base_type hang
E-15	hir_def, hir_ty, hir_lower	N-D arrays
E-16/17/22/40	hir_ty, hir_lower	\$table_model: 1-D, N-D multilinear, cubic spline, varargs sigs
E-18/20/23/32	hir_ty, hir_lower, basedb	arrays in analog functions: in/out/return/literals + array case
E-19	tokens→hir_lower	do ... while
E-21/44	parser, elaborate, hir_def, sim_back	paramset + hidden system parameters
E-24/55	hir_lower, mir, osdi	\$discontinuity; \$finish/\$stop/\$fatal honored
E-25	hir_ty, osdi stdlib	\$simparam\$str
E-26/51	osdi, sim_back	ac_stim: crash fix, then full AC-RHS injection
E-27/28/52	hir_lower, sim_back	the idt family: modulo, IC, assert/reset
E-29/36	sim_back	port-flow probes; probe-only branches
E-30	hir_ty, hir_lower	variadic analysis()
E-31	hir_lower	complex poles/zeros in root forms
E-32	osdi inst_data	integer persistent state (crash fix)
E-34	tokens→hir_lower	{...} concat + {n{...}} replication
E-35	lexer	EOF comment hang
E-37/38	hir_lower, mir_opt, parser	operator + precedence audits (~, >> fold, % level)
E-39	parser, hir_ty, osdi	derived natures unwired at the node-kind boundary
E-41/49/58/59	elaborate	implicit nets; hierarchical names; defparam; port concat + OR lists
E-42/54	sim_back noise, mir_autodiff, mir_llvm, osdi	correlated + node-free complex noise factors
E-43	hir_def (+diagnostics), hir_ty	initializers completed; arity diagnostics
E-45	osdi, hir_def, elaborate	nodesets + nature-attribute access (ABI 0.5)
E-46/48	lexer, syntax	escaped ids, based literals, single-pass unescaper
E-47	preprocessor, hir_ty, hir_lower	`default_transition + transition fixes
E-50	hir_ty validation	domain-binding validation
E-53	hir_lower, osdi eval	@(final_step) + phase lists
E-56	hir_ty, osdi setup	parameter defaults exempt from ranges
E-59	tokens→hir_lower, hir_ty	\$realtime; OR lists; recursion diagnostics
E-61	hir_lower	slew sign fix (operator-args audit)

Enhancement	Pipeline areas	One line
E-65	preprocessor	macro-recursion guard
E-66	13 crates	zero-warning cleanup (44 → 0)
E-67	tokens, parser, syntax, elaborate	generate audit: genvar fix, nesting, if/case
E-68	tests, mini_harness	integration suite enabled; VACASK removed
E-70	hir_ty validation	loop-context diagnostics
E-71	hir_ty fmt, hir_lower fmt, osdi	display-format surface + %b segfault
E-78	lexer→hir_lower, hir_ty	casex/casez don't-care masks
E-84	parser, elaborate, hir_ty, hir_lower, mir_opt, driver	LRM example sweep: 6 defect fixes (port-branch panic, parser robustness, silent undefined modules, \$port_connected on open ports, dead-op codegen, exit codes)
E-85	parser, elaborate, hir_def, hir_ty	`__FILE__`/`__LINE__` + connection part-selects (the last two sweep findings)
E-86	parser, elaborate, sim_back	hierarchical branch probes + 2 DAE fixes (V-source-to-internal open circuit, collapse-of-probed-branch)

Enhancements not listed (57, 60, 62–64, 69, 72–77, 79–83) changed no compiler sources — they were validation suites, documentation, benchmark tooling, or ngspice-side work (see the [ngspice report](#)).